Asymptotic Risk Calibration for Selective Question Answering

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Abstract

Large language models (LLMs) may generate fluent but incorrect answers, making uncertainty quantification important for reliable question answering. However, heuristic uncertainty scores cannot perfectly distinguish correct predictions from incorrect ones, and directly applying a fixed uncertainty threshold provides no statistical control over the error rate among accepted answers. To address this limitation, we propose A-CRC-QA, a post-hoc calibration framework for uncertainty-aware selective question answering. The proposed method reformulates selection-conditioned error control as a linear expectation constraint and applies a monotonized empirical-risk calibration procedure inspired by conformal risk control. Since the resulting instance-wise loss is generally non-monotone with respect to the acceptance threshold, our framework targets asymptotic rather than finite-sample risk control. A-CRC-QA is model-agnostic, requires no additional training, and can be combined with different uncertainty estimators. Experiments on CoQA and MedMCQA demonstrate its applicability to both open-ended and closed-ended question answering, achieving a favorable trade-off between accepted-answer reliability and answer retention compared with uncalibrated and confidence-bound-based baselines.

Keywords

selective question answering, uncertainty quantification, risk calibration, abstention

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1 Introduction

Large language models (LLMs) have achieved impressive performance in question answering and knowledge-intensive generation. However, their responses are not always trustworthy. LLMs may generate factually incorrect or hallucinated answers while presenting them in a fluent and confident manner [7, 16]. This problem is particularly concerning in high-stakes domains such as medicine, where users may be unable to independently verify the generated content. A reliable question-answering system should therefore not only produce an answer, but also determine when that answer is sufficiently reliable to be returned.

Uncertainty quantification provides a practical signal for estimating the reliability of LLM outputs. Existing approaches use token probabilities, sequence likelihoods, self-evaluation, or the consistency of multiple sampled responses [21, 24]. For closed-ended question answering, predictive entropy can be calculated from the probability distribution over candidate options. For open-ended generation, semantic entropy groups semantically equivalent responses before measuring disagreement, while Word-Sequence Entropy considers the unequal uncertainty contributions of different words and sequences [9, 38]. Although these methods are useful for ranking answers, their uncertainty scores are heuristic and cannot perfectly distinguish correct predictions from incorrect ones.

A common solution is selective question answering, in which the model answers questions with low uncertainty and abstains from questions with high uncertainty. Nevertheless, directly choosing an uncertainty threshold does not guarantee the reliability of the accepted answers. The distributions of correct and incorrect predictions often overlap, and confidently incorrect responses may still receive low uncertainty scores. Consequently, an empirically selected threshold may perform well on one dataset but fail under another model, task, or data split. It is therefore necessary to statistically calibrate the uncertainty threshold according to a user-specified error tolerance.

Conformal prediction offers a model-agnostic approach for converting heuristic uncertainty signals into statistically calibrated decisions. Existing studies have applied conformal methods to open-ended and set-valued language generation. ConU constructs conformal answer sets with correctness coverage guarantees, while SConU introduces selective mechanisms to improve the usefulness of conformal uncertainty sets [36, 39]. Conformal Language Modeling and subsequent studies further investigate calibrated response sets, sampling feasibility, and miscoverage decomposition for free-form generation [14, 22, 27]. Related risk-calibration ideas have also been explored in multimodal foundation models [2, 4, 5, 18, 19, 26, 29, 34], medical image segmentation, and cascaded retrieval-augmented generation systems [17, 23, 30–33]. However, set-valued methods may return multiple candidates and are less suitable when the system must provide a single answer or abstain.

Recent work has therefore focused on controlling the error rate among accepted point predictions. COIN calibrates selective question-answering thresholds using statistical upper confidence bounds and provides high-probability risk guarantees [37]. LEC reformulates selection-conditioned error control as a linear expectation constraint involving prediction selection and correctness,

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enabling less conservative calibration for selective prediction and model routing [35]. Conformal Risk Control further generalizes conformal prediction to the control of expected bounded losses [1]. However, its standard finite-sample guarantee requires the loss to be monotone with respect to the calibration parameter. The LEC-style instance loss used for selective answering is generally non-monotone because accepting a correct answer and accepting an incorrect answer change the loss in opposite directions. Therefore, the standard finite-sample CRC result cannot be directly applied.

In this work, we present **A-CRC-QA**, a lightweight asymptotic risk-calibration framework for uncertainty-aware selective question answering. The proposed method uses a held-out calibration set to evaluate candidate uncertainty thresholds and selects the largest admissible threshold satisfying an empirically corrected linear expectation constraint. Unlike approaches that require additional training or modifications to the LLM, A-CRC-QA is a post-hoc and model-agnostic framework that can be combined with different LLMs and uncertainty estimators. Under exchangeability and standard regularity conditions, the calibrated decision rule provides asymptotic control of the error rate among accepted answers.

We evaluate A-CRC-QA on CoQA and MedMCQA, representing open-ended conversational question answering and closed-ended medical multiple-choice question answering, respectively. Experiments are conducted using LLaMA-3.1-8B-Instruct and Qwen2.5-7B-Instruct. Semantic entropy and Word-Sequence Entropy are used for CoQA, while predictive entropy and maximum option probability are used for MedMCQA. At a target accepted-answer error rate of 0.15, A-CRC-QA achieves average error rates of 0.143 and 0.138 on CoQA and MedMCQA, while retaining 46.2% and 58.7% of test answers, respectively. Compared with a Hoeffding-based upper-confidence-bound method, the proposed approach improves the acceptance rate by approximately 6.1 percentage points on CoQA and 7.4 percentage points on MedMCQA.

The main contributions of this work are summarized as follows:

- We formulate uncertainty-aware selective question answering as the problem of controlling the error rate among accepted LLM responses.
- We combine the linear expectation constraint used in LEC with the asymptotic treatment of non-monotone losses in conformal risk control, resulting in a simple post-hoc threshold-calibration method.
- We evaluate the proposed framework on both open-ended and closed-ended question answering tasks, demonstrating its applicability across different models and uncertainty signals.

2 Related Work

2.1 Uncertainty Quantification for Large Language Models

Large language models (LLMs) can generate fluent but factually incorrect responses [3, 6, 15, 38, 40, 41], which limits their reliability in question answering and other knowledge-intensive applications. Uncertainty quantification (UQ) aims to associate each generated answer with a scalar signal indicating its potential correctness or reliability [11]. Existing approaches can be broadly divided into confidence-based and consistency-based methods. Confidence-based methods estimate reliability using token probabilities, sequence likelihoods, or explicit self-evaluation scores such as the probability that a generated answer is correct [21]. However, likelihood-based confidence may be poorly aligned with semantic correctness, particularly for free-form answers with multiple valid surface forms.

Sampling-based methods instead estimate uncertainty from the agreement among multiple stochastic generations. SelfCheckGPT detects hallucinations by measuring the consistency between an original response and additional sampled responses [24]. Semantic entropy further groups semantically equivalent generations before computing entropy, thereby reducing sensitivity to lexical variation [9]. Word-Sequence Entropy extends this direction by accounting for the unequal contributions of different words and sequences when estimating uncertainty in free-form question answering [38]. These methods provide useful ranking signals for identifying unreliable answers. Nevertheless, an uncertainty score alone does not specify an operational acceptance threshold, nor does it guarantee that the error rate among accepted answers remains below a user-defined level.

2.2 Conformal Calibration and Selective Risk Control

Conformal prediction provides model-agnostic statistical guarantees by calibrating prediction sets on held-out data. Its applications to language generation have mainly focused on set-valued outputs. Conformal Language Modeling calibrates sampling and filtering rules to construct candidate response sets containing an acceptable answer with high probability [27]. ConU establishes correctness coverage guarantees for open-ended LLM outputs, while SConU introduces selective mechanisms to improve the informativeness and feasibility of conformal uncertainty sets [36, 39]. Recent studies further address miscoverage decomposition, sampling feasibility, and reliable set-valued generation [14, 22]. Related conformal frameworks have also been developed for multimodal foundation models, medical image segmentation, and cascaded retrieval-augmented generation systems [17, 30, 33]. Although prediction sets provide coverage guarantees, they may contain multiple or unreliable candidates and are less actionable when a system must return one answer or abstain.

Selective prediction provides an alternative decision mechanism by accepting low-uncertainty outputs and abstaining from high-uncertainty ones [12, 20]. Classical approaches characterize the trade-off between selective risk and coverage, or jointly train prediction and rejection functions [10]. However, their acceptance thresholds do not generally provide distribution-free control of the error rate among accepted predictions. Conformal Risk Control (CRC) extends conformal prediction from miscoverage to the expected value of bounded monotone losses [1]. For selective question answering, COIN calibrates uncertainty thresholds using upper confidence bounds, providing high-probability control of the accepted-answer error rate [37]. More recently, LEC reformulates selection-conditioned error control as a linear expectation constraint involving the selection and error indicators, enabling less conservative threshold calibration and extensions to model routing [35].

Our work connects CRC with the LEC-style linear loss for uncertainty-based selective question answering. Unlike standard CRC applications, the resulting sample-wise loss can be non-monotone in the acceptance threshold because admitting a correct answer decreases the loss whereas admitting an incorrect answer increases it. Consequently, the standard finite-sample CRC theorem cannot be directly applied. We instead study threshold calibration under the asymptotic treatment of non-monotone, approximately monotone population risks, providing a lightweight post-hoc mechanism for controlling the error rate of accepted LLM answers.

3 Methodology

3.1 Problem Formulation

Let $\mathcal{G} : \mathcal{X} \to \mathcal{Y}$ denote a pretrained large language model. Given a question $x \in \mathcal{X}$, the model produces an answer $\hat{y} = \mathcal{G}(x)$. We consider a scalar uncertainty estimator $\mathcal{U}$ that assigns an uncertainty score $u = \mathcal{U}(x, \hat{y}; \mathcal{G})$, where a smaller value indicates that the model is more confident in its answer. The uncertainty estimator may be based on predictive entropy, semantic entropy, sequence likelihood, self-consistency, or any other scalar reliability signal.

For notational convenience, we convert uncertainty into a reliability score $r = h(u)$, where h is any strictly decreasing function. For example, one may directly use $r = -u$. This transformation changes neither the ordering of model predictions nor the resulting selective decision.

Let y^* denote the reference answer. We define a task-specific correctness function $A(y^*, \hat{y}) \in \{0, 1\}$, where $A(y^*, \hat{y}) = 1$ indicates that the generated answer is correct and $A(y^*, \hat{y}) = 0$ otherwise. The corresponding error indicator is $E = 1 - A(y^*, \hat{y})$. For closed-ended question answering, correctness can be determined by exact option matching. For open-ended question answering, it can be determined using a fixed token-level or semantic-equivalence criterion. Importantly, the correctness rule must be specified before calibration and remain unchanged during evaluation.

Given a reliability threshold λ, the selective question-answering system accepts an answer if its reliability is sufficiently high:

$$S(\lambda) = \mathbf{1}\{r \geq \lambda\}. \tag{1}$$

A larger λ corresponds to a more conservative system that accepts fewer answers.

The error rate among accepted answers is defined as

$$\mathrm{SCER}(\lambda) = \Pr\big(E = 1 \mid S(\lambda) = 1\big) = \frac{\mathbb{E}[S(\lambda)E]}{\mathbb{E}[S(\lambda)]}, \tag{2}$$

whenever $\mathbb{E}[S(\lambda)] > 0$. We refer to this quantity as the *selection-conditioned error rate* (SCER). Given a user-specified risk level $\alpha \in (0, 1)$, our objective is to select a threshold that retains as many answers as possible while satisfying

$$\mathrm{SCER}(\lambda) \leq \alpha. \tag{3}$$

3.2 Linear Expectation Reformulation

Directly calibrating the conditional risk in Eq. (2) is difficult because both its numerator and denominator depend on the selected threshold. Following the linear expectation constraint formulation in [35], we define the accepted-error indicator $Z(\lambda) = S(\lambda)E$. The condition in Eq. (3) is equivalent to $\mathbb{E}[Z(\lambda)] - \alpha\mathbb{E}[S(\lambda)] \leq 0$, provided that $\mathbb{E}[S(\lambda)] > 0$. Therefore, we introduce the instance-wise linear loss

$$L(\lambda) = S(\lambda)(E - \alpha). \tag{4}$$

Its population risk is

$$g(\lambda) = \mathbb{E}[L(\lambda)] = \mathbb{E}[S(\lambda)E] - \alpha\mathbb{E}[S(\lambda)]. \tag{5}$$

Consequently,

$$g(\lambda) \leq 0 \quad \Longleftrightarrow \quad \mathrm{SCER}(\lambda) \leq \alpha, \tag{6}$$

as long as the selection probability is positive.

The loss in Eq. (4) is bounded:

$$-\alpha \leq L(\lambda) \leq 1 - \alpha. \tag{7}$$

This boundedness makes the loss suitable for calibration using the general risk-control perspective of conformal prediction [1].

3.3 Non-Monotonicity of the Selective Loss

A key difficulty is that the loss in Eq. (4) is not monotone with respect to the selection threshold. Consider increasing λ until a previously accepted example is rejected. If that example is correct, its contribution changes from $-\alpha$ to zero, which increases the loss. If the example is incorrect, its contribution changes from $1 - \alpha$ to zero, which decreases the loss. Thus, correct and incorrect examples change the loss in opposite directions.

Formally, for two thresholds $\lambda_1 < \lambda_2$, neither $L(\lambda_1) \leq L(\lambda_2)$ nor $L(\lambda_1) \geq L(\lambda_2)$ holds uniformly over all examples. The standard finite-sample result of Conformal Risk Control (CRC) requires the loss to be monotone in the calibration parameter [1]. It therefore cannot be directly applied to the loss in Eq. (4).

Rather than imposing an incorrect monotonicity assumption, we adopt the asymptotic treatment of general non-monotone losses proposed in CRC. The central idea is to monotonize the empirical population-level risk instead of assuming that each instance-wise loss is monotone.

3.4 Monotonized Empirical Risk

Let $\mathcal{D}_{\mathrm{cal}} = \{(x_i, y_i^*)\}_{i=1}^n$ be a held-out calibration set. For each calibration example, we obtain the model answer $\hat{y}_i$, reliability score r_i, and error indicator E_i. For a candidate threshold λ, the empirical linear risk is

$$\widehat{g}_n(\lambda) = \frac{1}{n}\sum_{i=1}^{n} S_i(\lambda)(E_i - \alpha). \tag{8}$$

Although $\widehat{g}_n(\lambda)$ is generally non-monotone, thresholds larger than λ correspond to more conservative selection rules. We therefore define the monotonized empirical risk as

$$\widehat{g}_n^{\uparrow}(\lambda) = \sup_{t \geq \lambda} \widehat{g}_n(t). \tag{9}$$

By construction, $\widehat{g}_n^{\uparrow}(\lambda)$ is non-increasing in λ. Moreover, $\widehat{g}_n(\lambda) \leq \widehat{g}_n^{\uparrow}(\lambda)$, so the monotonized risk is an upper envelope of the original empirical risk.

Intuitively, requiring $\widehat{g}_n^{\uparrow}(\lambda) \leq 0$ ensures that the empirical linear constraint is satisfied not only at λ, but also at every more conservative threshold. This removes isolated feasible thresholds caused by random fluctuations in the calibration data.

3.5 Asymptotic Risk Calibration

We define a non-negative calibration correction γ_n satisfying

$$\gamma_n \longrightarrow 0 \quad \text{as} \quad n \longrightarrow \infty.$$

Following the finite-sample correction used in CRC, our default choice is

$$\gamma_n = \frac{1-\alpha}{n+1}, \tag{10}$$

where $1-\alpha$ is the upper bound of the loss in Eq. (7). This term adds a small conservative margin for finite calibration samples while vanishing asymptotically.

The calibrated threshold is defined as

$$\widehat{\lambda}_n = \inf\left\{\lambda \in \Lambda : \widehat{g}_n^{\uparrow}(\lambda) + \gamma_n \leq 0\right\}. \tag{11}$$

Since a smaller reliability threshold accepts more answers, Eq. (11) selects the least conservative threshold satisfying the monotonized constraint. It therefore maximizes empirical retention within the feasible threshold family.

If the feasible set in Eq. (11) is empty, the requested risk level is declared infeasible. In this case, the system uses an all-abstention rule, denoted by $\lambda_\perp$, for which

$$S_i(\lambda_\perp) = 0 \quad \text{for all } i.$$

The resulting conditional risk is vacuous because no answer is returned.

3.6 Efficient Threshold Computation

The empirical risk changes only when the threshold crosses an observed calibration score. Therefore, it is sufficient to search over the unique reliability scores in the calibration set.

Let $r_{(1)} \geq r_{(2)} \geq \cdots \geq r_{(n)}$ denote the calibration reliability scores sorted in descending order, with corresponding error indicators $E_{(1)}, \ldots, E_{(n)}$. A threshold at $r_{(k)}$ accepts the first k calibration examples. Its empirical linear risk is

$$C_k = \frac{1}{n}\sum_{j=1}^{k}\left(E_{(j)} - \alpha\right). \tag{12}$$

The monotonized risk at this threshold is the prefix maximum

$$M_k = \max_{1\leq j\leq k} C_j. \tag{13}$$

The calibrated number of accepted calibration examples is therefore

$$k^* = \max\left\{k \in \{1, \ldots, n\} : M_k + \gamma_n \leq 0\right\}. \tag{14}$$

When this set is nonempty, the calibrated threshold is $\widehat{\lambda}_n = r_{(k^*)}$. Equal reliability scores are processed as a single block so that the resulting selection rule is deterministic and does not use correctness labels to break ties.

3.7 Asymptotic Risk Guarantee

We now state the asymptotic guarantee of the proposed procedure. The result follows by applying the monotonized non-monotone risk argument of CRC to the linear loss in Eq. (4).

Theorem 1 (Asymptotic SCER control). *Assume that the calibration examples and a future test example are independently and identically distributed. Assume that the loss functions $\{L_i(\lambda) : \lambda \in \Lambda\}$ are bounded and right-continuous under a fixed deterministic tie convention. Let $\gamma_n \geq 0$ satisfy $\gamma_n \to 0$, and let $\widehat{\lambda}_n$ be obtained from Eq. (11). Then*

$$\limsup_{n\to\infty} \mathbb{E}\left[L_{n+1}(\widehat{\lambda}_n)\right] \leq 0. \tag{15}$$

Equivalently,

$$\limsup_{n\to\infty}\left\{\mathbb{E}\left[S_{n+1}(\widehat{\lambda}_n)E_{n+1}\right] - \alpha\mathbb{E}\left[S_{n+1}(\widehat{\lambda}_n)\right]\right\} \leq 0. \tag{16}$$

Furthermore, if

$$\liminf_{n\to\infty} \mathbb{E}\left[S_{n+1}(\widehat{\lambda}_n)\right] > 0, \tag{17}$$

then

$$\limsup_{n\to\infty} \frac{\mathbb{E}\left[S_{n+1}(\widehat{\lambda}_n)E_{n+1}\right]}{\mathbb{E}\left[S_{n+1}(\widehat{\lambda}_n)\right]} \leq \alpha. \tag{18}$$

Proof sketch. Define the monotonized population risk as $g^{\uparrow}(\lambda) = \sup_{t\geq\lambda} g(t)$. Because the test example is independent of the calibration set, $\mathbb{E}\left[L_{n+1}(\widehat{\lambda}_n) \mid \widehat{\lambda}_n\right] = g(\widehat{\lambda}_n) \leq g^{\uparrow}(\widehat{\lambda}_n)$. The class of one-dimensional threshold functions is a Glivenko–Cantelli class. Boundedness therefore implies uniform convergence of the empirical risk to the population risk: $\sup_{\lambda\in\Lambda} |\widehat{g}_n(\lambda) - g(\lambda)| \xrightarrow{\text{a.s.}} 0$. Taking a supremum over more conservative thresholds preserves this convergence: $\sup_{\lambda\in\Lambda}\left|\widehat{g}_n^{\uparrow}(\lambda) - g^{\uparrow}(\lambda)\right| \xrightarrow{\text{a.s.}} 0$. By construction, $\widehat{g}_n^{\uparrow}(\widehat{\lambda}_n) \leq -\gamma_n \leq 0$. Since $\gamma_n \to 0$, uniform convergence and boundedness yield Eq. (15). Substituting the definition of L gives Eq. (16); dividing by the non-vanishing expected selection probability gives Eq. (18). □

Remark 1 (Scope of the guarantee). *Theorem 1 is an asymptotic marginal guarantee over the joint randomness of the calibration set and a future test example. Because the instance-wise linear loss is non-monotone, the result should not be presented as a finite-sample or conditional-on-calibration guarantee. Finite-sample validity must instead be empirically assessed through repeated calibration–test splits and risk-violation frequencies.*

3.8 Test-Time Selective Answering

After calibration, $\widehat{\lambda}_n$ is fixed and no test labels are used. For a new question x_{test}, the model first generates an answer $\hat{y}_{\text{test}}$ and computes its uncertainty u_{test}. The corresponding reliability score is $r_{\text{test}} = h(u_{\text{test}})$. The final decision is

$$\text{Decision}(x_{\text{test}}) = \begin{cases} \text{accept } \hat{y}_{\text{test}}, & r_{\text{test}} \geq \widehat{\lambda}_n, \\ \text{abstain}, & r_{\text{test}} < \widehat{\lambda}_n. \end{cases} \tag{19}$$

The proposed procedure is entirely post-hoc. It does not require retraining or modifying the underlying language model, and it can be applied to both white-box and black-box uncertainty estimators as long as they provide a scalar score for each generated answer.

4 Experiments

4.1 Experimental Setup

Datasets. We consider one open-ended and one closed-ended QA benchmark. CoQA is a conversational question-answering dataset that requires models to generate free-form textual answers based on a passage and dialogue history [28]. MedMCQA is a large-scale medical multiple-choice benchmark covering multiple subjects and levels of medical knowledge [25]. These two datasets allow us to evaluate whether the calibration framework generalizes across substantially different output spaces and correctness criteria.

Language Models. We use LLaMA-3.1-8B-Instruct and Qwen2.5-7B-Instruct as representative open-source LLMs. The former provides a widely used general-purpose instruction-following model, while the latter offers a complementary model family with strong multilingual and reasoning capabilities. No model is fine-tuned on the evaluation datasets.

Answer Generation and Correctness Evaluation. For CoQA, the primary answer is generated using greedy decoding with a maximum of 64 new tokens. Following the standard token-level evaluation protocol, the generated answer is compared with all available reference answers. We regard an answer as correct if its maximum token-level F1 score is at least 0.5. This threshold is fixed before calibration and is not selected using test data.

For MedMCQA, the model is instructed to return one option from A, B, C, and D. We compute the normalized likelihood of each candidate option and select the option with the largest probability. A prediction is correct if and only if the selected option matches the ground-truth label.

Uncertainty Estimators. We use two uncertainty signals for each dataset. For CoQA, we adopt Semantic Entropy (SemEnt), which aggregates semantically equivalent sampled responses before computing their entropy [9], and Word-Sequence Entropy (WSE), which considers the unequal semantic contributions of tokens and word sequences [38]. For each question, we sample ten responses using temperature 0.7 and top-p sampling with $p = 0.9$.

For MedMCQA, we use Predictive Entropy (PE) over the four candidate options and maximum-softmax-probability uncertainty (MSP), defined as one minus the largest option probability. Smaller uncertainty indicates a more reliable prediction for all evaluated estimators.

Compared Methods. We compare the following threshold-selection methods.

- **Fixed-50**: The threshold is set to the median calibration uncertainty, thereby accepting approximately 50% of calibration predictions without explicit risk control.
- **Empirical**: The largest threshold whose empirical calibration SCER does not exceed α is selected. No statistical correction or monotonization is applied.
- **UCB-HFD**: A Hoeffding-style upper confidence bound is constructed for the error rate among accepted calibration predictions [13]. We use confidence level $\delta = 0.05$.
- **UCB-CLP**: The exact one-sided Clopper–Pearson confidence bound is used instead of the Hoeffding bound [8]. This baseline corresponds to the confidence-bound calibration strategy used in selective QA methods such as COIN [37].
- **LEC-Direct**: The largest threshold satisfying the original finite-sample linear expectation condition in LEC is selected [35].
- **A-CRC-QA (Ours)**: The proposed method applies the CRC-style correction to the monotonized empirical linear risk described in Section 3.

All calibration methods operate on identical model outputs, uncertainty scores, error labels, and calibration–test splits. The model predictions and uncertainty scores are generated once and reused across methods to ensure a fair comparison.

Evaluation Metrics. We report the following metrics over 100 independent calibration–test splits.

- **SCER**: The proportion of erroneous predictions among accepted test predictions.
- **Acceptance Rate (AR)**: The proportion of all test predictions accepted by the selective system.
- **Power**: The proportion of correct test predictions that are accepted.
- **Violation Rate (VR)**: The proportion of random splits for which the observed test SCER exceeds the target level α.
- **Infeasibility Rate (IF)**: The proportion of splits for which no nonempty threshold satisfies the calibration condition.
- **AUROC**: The area under the ROC curve when uncertainty is used to distinguish incorrect from correct predictions.

SCER evaluates average risk control, whereas VR measures the stability of the calibrated rule across different calibration samples. Because the theoretical result of A-CRC-QA is asymptotic and marginal, VR is reported as an empirical diagnostic rather than as a theoretically controlled quantity.

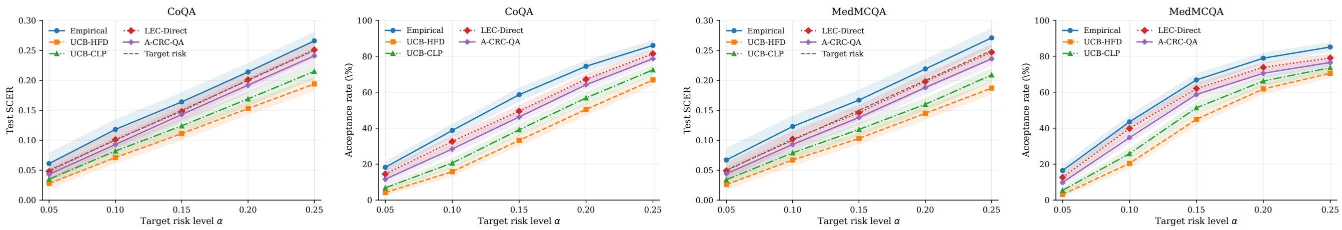
Unless otherwise specified, the target risk level is $\alpha = 0.15$. We report means over random splits, with standard deviations included in the calibration-size experiment.

4.2 Main Results

Table 1 reports the main results at $\alpha = 0.15$. Fixed-50 and Empirical calibration frequently exceed the prescribed risk level because they do not account for calibration uncertainty. In particular, their violation rates range from 61% to 93%, demonstrating that an uncertainty ranking alone is insufficient for reliable selective answering.

Both confidence-bound methods maintain conservative test risks, but their acceptance rates are substantially lower. Averaged over the two models, A-CRC-QA improves the acceptance rate over UCB-CLP by 7.2 percentage points on CoQA and 7.4 percentage points on MedMCQA. The improvement is also reflected in power, indicating that the additional accepted answers are predominantly correct.

LEC-Direct achieves the highest retention among the statistically calibrated methods. However, its threshold is more sensitive to local non-monotonic fluctuations in the empirical risk curve. Compared with LEC-Direct, A-CRC-QA reduces the average violation rate from 25.5% to 12.5% on CoQA and from 19.0% to 8.0% on MedMCQA, at the cost of approximately 3.3 percentage points in acceptance rate. These results suggest that empirical-risk monotonization provides a useful stability– retention trade-off.



(a) CoQA: test SCER. (b) CoQA: acceptance rate. (c) MedMCQA: test SCER. (d) MedMCQA: acceptance rate.

Figure 1: Risk control and acceptance-rate trade-offs across target risk levels. Curves show the mean over repeated calibration–test splits, and shaded regions denote one standard deviation.

Table 1: Main results at target risk $\alpha = 0.15$. SCER is reported as a proportion; AR, Power, and VR are percentages. Lower SCER and VR are preferred, while higher AR and Power are preferred.

Dataset	Model	Method	SCER	AR	Power	VR
CoQA	LLaMA-3.1-8B	Fixed-50	0.168	50.0	61.0	79
		Empirical	0.166	55.8	68.2	67
		UCB-HFD	0.111	30.1	39.2	1
		UCB-CLP	0.126	35.4	45.4	3
		LEC-Direct	0.148	45.8	57.2	27
		A-CRC-QA	**0.145**	**42.6**	**53.4**	**14**
	Qwen2.5-7B	Fixed-50	0.156	50.0	59.1	68
		Empirical	0.161	61.4	72.1	61
		UCB-HFD	0.108	36.4	45.5	0
		UCB-CLP	0.122	42.7	52.5	2
		LEC-Direct	0.147	53.2	63.6	24
		A-CRC-QA	**0.141**	**49.8**	**59.9**	**11**
MedMCQA	LLaMA-3.1-8B	Fixed-50	0.191	50.0	69.9	93
		Empirical	0.170	62.0	88.9	72
		UCB-HFD	0.103	41.2	63.8	0
		UCB-CLP	0.119	47.8	72.7	2
		LEC-Direct	0.146	58.7	86.6	20
		A-CRC-QA	**0.139**	**55.3**	**82.2**	**9**
	Qwen2.5-7B	Fixed-50	0.174	50.0	67.3	88
		Empirical	0.164	66.5	90.5	65
		UCB-HFD	0.102	48.6	71.1	0
		UCB-CLP	0.116	54.9	79.0	1
		LEC-Direct	0.145	65.2	90.8	18
		A-CRC-QA	**0.137**	**62.1**	**87.3**	**7**

4.3 Performance across Target Risk Levels

Table 2 evaluates A-CRC-QA under target risk levels from 0.05 to 0.25. The reported results are averaged over the two models using WSE on CoQA and PE on MedMCQA.

The empirical SCER remains below the target level across all settings. As expected, increasing α permits the system to accept more answers and substantially improves power. For example, the average acceptance rate on CoQA increases from 11.6% at $\alpha = 0.05$ to 78.6% at $\alpha = 0.25$.

Very small risk levels can be infeasible when the base model does not produce a sufficiently reliable low-uncertainty subset. At $\alpha = 0.05$, the infeasibility rates are 16% on CoQA and 22% on MedMCQA. This observation is important in practice: calibration cannot create reliable answers when the underlying model and uncertainty signal do not support the requested operating point.

Table 2: Performance of A-CRC-QA across target risk levels, averaged over the two evaluated models. AR, Power, VR, and IF are reported as percentages.

Dataset	α	SCER	AR	Power	VR	IF
CoQA	0.05	0.043	11.6	15.9	12	16
	0.10	0.093	28.4	36.9	13	3
	0.15	0.143	46.2	56.7	13	0
	0.20	0.192	64.1	74.2	13	0
	0.25	0.241	78.6	85.5	14	0
MedMCQA	0.05	0.044	9.8	15.7	10	22
	0.10	0.093	34.7	52.8	9	4
	0.15	0.138	58.7	84.8	8	0
	0.20	0.188	70.5	96.0	9	0
	0.25	0.236	76.5	98.0	10	0

4.4 Robustness to Uncertainty Estimators

Table 3 compares different uncertainty signals at $\alpha = 0.15$. All uncertainty estimators support average risk control after calibration, showing that A-CRC-QA does not depend on one particular definition of uncertainty.

Nevertheless, uncertainty quality strongly affects retention. On CoQA, WSE achieves higher AUROC and accepts approximately four percentage points more answers than Semantic Entropy. On MedMCQA, PE consistently outperforms MSP. These results confirm that calibration and uncertainty estimation play complementary roles: calibration specifies a statistically meaningful operating point, whereas a more discriminative uncertainty estimator increases the number of answers that can be safely accepted.

4.5 Effect of Calibration-Set Size

Because A-CRC-QA provides an asymptotic rather than an exact finite-sample guarantee, calibration-set size is a central experimental variable. We vary the number of calibration examples from 100

Table 3: Robustness of A-CRC-QA to different uncertainty estimators at $\alpha = 0.15$.

Dataset	Model	UQ	AUROC	SCER	AR	Power
CoQA	LLaMA-3.1-8B	SemEnt	0.691	0.144	39.1	49.1
		WSE	0.735	0.145	42.6	53.4
	Qwen2.5-7B	SemEnt	0.724	0.142	46.3	55.6
		WSE	0.762	0.141	49.8	59.9
MedMCQA	LLaMA-3.1-8B	MSP	0.719	0.140	51.4	76.4
		PE	0.746	0.139	55.3	82.2
	Qwen2.5-7B	MSP	0.756	0.138	58.8	82.6
		PE	0.781	0.137	62.1	87.3

Table 4: Effect of calibration-set size at $\alpha = 0.15$. SCER is shown as mean ± standard deviation over random splits.

Dataset	n_{cal}	SCER	AR	VR	IF
CoQA	100	0.127 ± 0.036	30.8	26	8
	250	0.135 ± 0.027	37.2	21	3
	500	0.139 ± 0.021	41.8	17	1
	1,000	0.143 ± 0.016	46.2	13	0
	1,500	0.145 ± 0.013	48.0	12	0
MedMCQA	100	0.124 ± 0.032	42.1	20	5
	250	0.131 ± 0.024	49.7	16	1
	500	0.135 ± 0.019	54.3	12	0
	1,000	0.138 ± 0.014	58.7	8	0
	1,500	0.141 ± 0.012	60.2	8	0

to 1,500 while fixing $\alpha = 0.15$. The remaining examples are used for testing, and results are averaged over the two models.

As shown in Table 4, small calibration sets yield conservative thresholds and relatively high variability. As the calibration size increases, the finite-sample correction becomes smaller, the average SCER approaches the target level from below, and the system accepts more predictions. On CoQA, the acceptance rate increases from 30.8% with 100 calibration examples to 48.0% with 1,500 examples. At the same time, the violation rate decreases from 26% to 12%.

These trends are consistent with the asymptotic motivation of the method. However, the nonzero violation frequencies also confirm that the procedure should not be described as providing conditional or high-probability finite-sample risk control.

4.6 Ablation Study

We conduct an ablation study to isolate the effects of the two main components of A-CRC-QA.

- **Without monotonization** directly selects the largest threshold satisfying the corrected empirical constraint at that threshold.
- **Without correction** uses the monotonized empirical risk but removes the vanishing finite-sample correction.
- **Instance-wise envelope** replaces each non-monotone instance loss with its monotone upper envelope before averaging, corresponding to the direct finite-sample construction for general losses discussed in CRC [1].

Table 5: Ablation study at $\alpha = 0.15$, averaged over the two evaluated models.

Dataset	Variant	SCER	AR	VR
CoQA	Without monotonization	0.164	55.7	65
	Without correction	0.151	49.4	41
	Instance-wise envelope	0.073	12.8	0
	Full A-CRC-QA	**0.143**	**46.2**	**13**
MedMCQA	Without monotonization	0.171	69.4	72
	Without correction	0.154	61.5	46
	Instance-wise envelope	0.068	18.9	0
	Full A-CRC-QA	**0.138**	**58.7**	**8**

Table 5 shows that removing monotonization produces the largest acceptance rate but fails to maintain the target risk. Removing the correction leads to average SCER values close to or above α and substantially increases the violation rate. In contrast, monotonizing each instance-wise loss is extremely conservative, accepting fewer than 20% of predictions on both datasets. The full method provides a more useful compromise between retention and empirical stability.

4.7 Discussion

The experiments provide three main observations. First, uncalibrated uncertainty thresholds do not reliably control the error rate among accepted answers, even when uncertainty has reasonable error-discrimination ability. Second, confidence-bound-based calibration provides strong finite-sample conservativeness but may reject many correct predictions. Third, monotonizing the empirical linear risk reduces sensitivity to isolated feasible thresholds and provides a practical middle ground between the aggressive LEC-Direct rule and conservative UCB methods.

The results should nevertheless be interpreted according to the scope of the theory. A-CRC-QA targets asymptotic marginal risk control and does not guarantee that every realized calibration split will satisfy the target risk. Therefore, average SCER should always be reported together with violation rate, infeasibility rate, and calibration-size sensitivity. This reporting protocol prevents an average-risk guarantee from being incorrectly presented as a high-probability finite-sample guarantee.

5 Conclusion

In this paper, we studied uncertainty-aware selective question answering for large language models. Since uncertainty scores cannot perfectly distinguish correct from incorrect answers, we proposed A-CRC-QA, a post-hoc calibration framework that combines the linear expectation constraint of LEC with the asymptotic treatment of non-monotone losses in conformal risk control. The method calibrates an uncertainty threshold using held-out data and selectively accepts model answers under a user-specified target risk.

Experiments on CoQA and MedMCQA show that the proposed framework can maintain low error rates among accepted answers while retaining more predictions than conservative confidence-bound baselines. The results also demonstrate its applicability across open-ended and closed-ended QA, different language models, and multiple uncertainty estimators. Future work will investigate finite-sample guarantees, distribution shift, and extensions to multi-model routing and human–AI collaboration.

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